

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.3: The Periodic Table — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: The Periodic Table
Driving Question	DRIVING QUESTION: Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Mystery Element Card Challenge — Launching the Phenomenon	Learners sorted 20 unlabelled element cards (H–Ca) — each showing a Kenyan context photograph — into self-chosen groups. Different groups produced different arrangements: some sorted by atomic number, some by reactivity rating, some by state at room temperature, and some by the Kenyan context shown on the card. Comparison of three element pairs (Li/Na, F/Cl, He/Ne) showed that elements far apart in atomic number can share very similar properties, while neighbouring elements can be completely different.	The 20 elements from Hydrogen to Calcium can be arranged into a two-dimensional grid of GROUPS (vertical columns) and PERIODS (horizontal rows). Elements in the same group share similar properties even when their atomic numbers are far apart. Scientists disagreed about how to organise the elements — just as our class disagreed — until patterns in behaviour made certain groupings more defensible than others. The unit Driving Question is: 'Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do?'	We cannot yet explain WHY elements in the same group behave similarly — that explanation requires understanding what is INSIDE atoms (electron arrangement), which we will build over the next lessons. Our initial models (Model Version 1) are our best current hypotheses. They are not wrong — they are waiting for evidence.
Lesson 2 History of the Periodic Table: Standing on the Shoulders of Giants	Students watched a curated video on the historical development of the periodic table (Döbereiner → Newlands → Mendeleev → Moseley) and completed a structured observation guide tracking each scientist's ordering method and its limitations. Groups constructed annotated timeline posters connecting each historical strategy to their own Lesson 1 card-sort approaches, and conducted a gallery walk with peer-	Students learned that the periodic table was not invented in a single moment but evolved over nearly 100 years as scientists proposed, tested, and revised different ordering systems. They learned that Mendeleev's use of atomic mass allowed brave predictive gaps, but that Moseley's use of atomic number — the number of protons — finally resolved all anomalies and revealed a deeper, invisible pattern	Students cannot yet fully explain WHY ordering by atomic number works — specifically, what internal feature of the atom (electron arrangement) causes elements in the same group to behave similarly. This gap is intentionally preserved in their Sentence 2 model annotation and on the DQB, and will be directly investigated in Lesson 3.

	challenge sticky notes.	inside the atom. They connected this historical arc to their own card-sort strategies from Lesson 1.	
Lesson 3 Electron Arrangement and Periodic Table Position	Students placed 20 element cards (H–Ca) onto a large floor grid (4 periods × 8 groups) after writing s/p electron configurations, and conducted a gallery walk to observe patterns across rows and columns.	The group number equals the number of outer-shell (valence) electrons, and the period number equals the number of occupied electron shells — the invisible internal structure that explains why Na and K behave alike, and why Cl (Gigiri water treatment) and Ar (welding) behave so differently despite being neighbours.	Elements in the same group share the same number of outer-shell electrons, which drives their similar chemical behaviour — the hidden rule that resolves the Mystery Element Card sorting puzzle introduced in Lesson 1.
Lesson 4 Chemical Families: Identifying Groups and Trends in the Periodic Table	Students sorted Chemical Family Data Cards into Groups 1, 2, 17, 18, and transition elements; recorded reactivity and physical property trends in a Trend Detective Table; and identified that Group 1 reactivity increases going down (Li → Na → K) while Group 17 reactivity decreases going down (F → Cl → Br).	Students learned the names and positions of the five major chemical families in the periodic table; that elements in the same group share similar properties due to their family membership; that reactivity trends within groups are directional and predictable; and that Kenyan materials such as potassium fertilisers (Rift Valley), chlorine for water treatment (Gigiri, Nairobi), and calcium in Maasai cattle bones are direct real-world expressions of these chemical families.	Students explained, using Claim-Evidence-Reasoning, that potassium is more reactive than lithium because reactivity increases down Group 1; they identified that the position of an element in the periodic table predicts its behaviour; and they acknowledged that their current model shows WHERE the trends occur but cannot yet explain WHY — pointing to electron arrangement as the next piece of evidence needed to fully answer the driving question.
Lesson 5 Ion Formation: Cations and Anions	When sodium and chlorine element cards were placed next to each other in our periodic table grid, they showed completely different reactivity ratings and contexts — sodium in table salt and chlorine as a disinfectant gas. We observed from dot-and-cross diagrams on the board that sodium ends up with 10 electrons after losing one, and chlorine ends up with 18 electrons after gaining one.	Atoms form ions by losing electrons (forming positively charged cations) or gaining electrons (forming negatively charged anions). The number of electrons lost or gained is determined by the element's group number and the drive toward a full outer shell described by the octet rule. Na forms Na ⁺ , K forms K ⁺ , Ca forms Ca ²⁺ , Cl forms Cl ⁻ , O forms O ²⁻ , and S forms S ²⁻ .	The element card for sodium showed high reactivity and a 1+ charge tendency, while chlorine showed a tendency to react as a 1- species — today we explained this using electron arrangement. Sodium is in Group 1 with one outer-shell electron and loses it to achieve the stable 2,8 configuration. Chlorine is in Group 17 with seven outer-shell electrons and gains one to achieve the stable 2,8,8 configuration. Their opposite electron-transfer directions explain why elements sitting next to each other in Period 3 behave so completely differently — answering part of the Lesson 1 puzzle from the card sort.
Lesson 6 Oxidation Numbers and Valency — Reading the Periodic Table's Charge Language	Students tabulated valencies and oxidation numbers for Groups 1, 2, 16, and 17 from electron arrangement data; examined photographs of FeCl ₂ and FeCl ₃ alongside images of rusted and clean iron at Kisumu Port; and completed a Radicals Reference	An element's group number in the periodic table predicts its fixed oxidation number (Group 1 → +1; Group 2 → +2; Group 16 → -2; Group 17 → -1). Transition elements such as iron have variable oxidation numbers (Fe ²⁺ and Fe ³⁺), which is why	The oxidation number of an element arises directly from the number of electrons it gains or loses to achieve a stable noble gas electron arrangement — connecting today's lesson back to the electron arrangement work in Lesson 3 and the ion formation

	Card linking SO_4^{2-} , NO_3^- , CO_3^{2-} , OH^- , and NH_4^+ to Kenyan contexts (Rift Valley fertiliser, Naivasha horticulture, Bamburi limestone, Bidco soap, DAP fertiliser).	rusted iron at Kisumu Port can appear different colours. Radicals are groups of atoms that carry a net charge and act as a single unit in chemical reactions; their valency equals the magnitude of their charge.	evidence in Lesson 5. Transition elements have incompletely filled d sub-shells that allow them to lose different numbers of electrons depending on conditions, explaining variable oxidation numbers. This means that an element's position in the periodic table encodes its charge language — a direct, partial answer to the driving question.
Lesson 7 Deriving Chemical Formulae Using Valency and the Crossover Method	Students observed the crossover method applied to calcium chloride and sodium chloride on the board, and observed how crossing the numerical values of ion charges generates the correct subscript ratio in the formula.	Students learned that the crossover method uses valency (the magnitude of the ion charge) to determine the subscript of each ion in an ionic formula, that brackets must enclose radicals when the subscript is greater than 1, and that the resulting formula can be verified by checking that total positive charge equals total negative charge.	The pattern is explained by the requirement for ionic compounds to be electrically neutral — the number of each ion type in the formula must be exactly the ratio that makes the sum of positive charges equal the sum of negative charges, which is precisely what the crossover method calculates from the valencies derived from periodic table position.
Lesson 8 Writing and Balancing Chemical Equations	Students observed that the atom inventory table reveals imbalances in symbol equations for real reactions including sodium reacting with water and calcium carbonate decomposing at the Athi River cement plant.	Students learned the three-step method for writing and balancing chemical equations: (1) write the word equation, (2) replace substances with correct formulae to give the unbalanced symbol equation, (3) adjust coefficients — never subscripts — using an atom inventory table until atoms are equal on both sides.	The requirement to balance equations is explained by the principle that atoms are conserved in chemical reactions — no atom is created or destroyed, only rearranged — which means the total number of each type of atom must be identical in the reactants and the products, connecting to the Law of Conservation of Mass.
Lesson 9 Consolidation and Extended Problem Solving: From Periodic Table Position to Balanced Equation	Students observed that urea, calcium ammonium nitrate, and muriate of potash each contain elements from identifiable chemical families on the periodic table (non-metals C, N, O; alkali metal K; alkaline earth metal Ca; halogen Cl). They also observed that the balanced equation for potassium reacting with water follows the same pattern as sodium reacting with water, consistent with both being in Group 1.	Students learned that the complete chain from periodic table position to balanced equation can be applied to real Kenyan agricultural chemicals. They confirmed that valency derived from electron arrangement correctly predicts compound formulae, and that balancing equations requires conservation of atoms on both sides (consistent with the Law of Conservation of Mass from Lesson 8).	Students explained that an element's position in the periodic table (its group) determines its number of outer electrons, which determines whether it forms a cation or anion and with what valency. That valency determines the formula of its ionic compounds via the crossover method, and those formulae are the building blocks of balanced chemical equations — all of which can be read from the element cards that seemed purely puzzling in Lesson 1.
Lesson 10 Model Gallery Walk, DQB Closure & Unit Reflection	Students observed (in Lesson 1) that element cards for sodium and chlorine looked completely different — different reactivity ratings, different Kenyan contexts (ugali salt vs. Gigiri water treatment), different physical appearances — yet by the end of the unit they discovered these two	Students learned that every element's behaviour is determined by its electron arrangement, which is encoded in its position on the periodic table (period = number of shells; group = number of valence electrons). This arrangement determines whether an atom forms a cation	The mystery of the element cards is resolved: elements that seem physically different can be chemically complementary because their electron arrangements are mirror-image incomplete — sodium has one electron to give, chlorine has one space to receive. The 'invisible rule' (electron

	elements combine to form the single most familiar compound in Kenyan daily life: the salt from the Malindi salt flats that seasons ugali across the country.	(loses electrons, e.g. $\text{Na} \rightarrow \text{Na}^+$) or an anion (gains electrons, e.g. $\text{Cl} \rightarrow \text{Cl}^-$), its valency and oxidation number, the formula of its compounds (using valency crossover), and the balanced equation for its reactions. The periodic table is therefore a predictive map, not merely a list.	arrangement seeking noble-gas stability) explains every visible pattern on the cards and on the periodic table. This same principle — that electrons determine bonding — will drive Sub-Strand 1.4, where students will investigate WHY Na^+ and Cl^- attract each other and what that attraction means for the physical properties of substances.
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